ROCK MAGNETISM AND THE EARTH'S MAGNETIC FIELD DURING PALEOZOIC TIME*

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(Received February 4, 1954)

ABSTRACT

A brief appraisal is given of the observations and arguments that are advanced in support of the opinion that during Paleozoic time the earth's magnetic field retained approximately its present orientation, and, except for possible brief excursions, its present sense.

Since 1945, continuous attention has been given to the subject of rock magnetism at the Department of Terrestrial Magnetism, Carnegie Institution of Washington. This program was initiated about 1935 and has been focused on the question, "To what extent is it possible to trace the history of the earth's magnetic field in geologic time by the study of the magnetic properties of rocks?". Results obtained up to 1949 have been reported in various papers. The present brief paper gives an abbreviated account of the results and conclusions reached during the past 3-1/2 years of almost uninterrupted work. It is our intention to publish the full details in the near future. During the course of this study, a number of individuals have participated: particularly grateful mention must be made of the contributions of O. W. Torreson and P. F. Michelsen, and of the helpful and stimulating criticism of M. A. Tuve and H. E. Tatel.

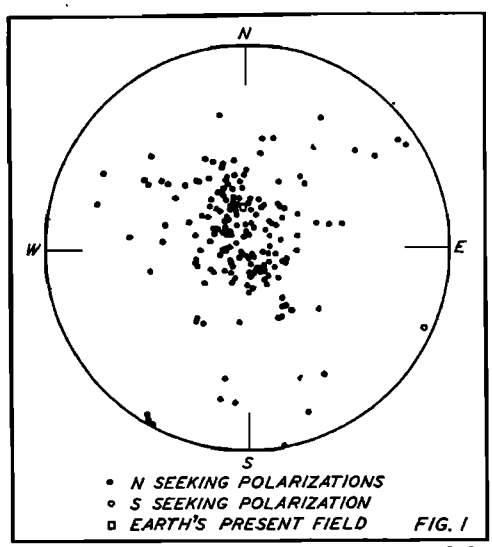
Simply stated, we think there is, in our data, a fair case for believing that since Cambrian time the earth's field has remained essentially in its present sense and roughly in its present orientation. The word "think" is used advisedly, for we do not have a proof that is certain, and the dikes, sills, and sediments having inverse magnetizations still are not completely understood. We are of the opinion, however, that anyone who concludes that the earth's field has reversed since Cambrian time is drawing a conclusion on questionable grounds.

As to the question of the degree of variability of the earth's total moment in past times, we as yet are not ready to say much. We have found a limited number of sediments having undetectable moments ($< 10^{-8}$ cgs/cc) and yet a respectable susceptibility, one that by comparison with other samples suggests that a reasonable residual moment should be present. Whether this property is an expression of a gradual demagnetization, or of unfavorable circumstances of deposition, or of a lack of the earth's field at the time of deposition, we do not know. As to the upper limit of intensity of the earth's field, we have in mind a promising method of approach which as yet we have examined in only a preliminary way. It depends

*Presented to an informal conference at the Institute of Geophysics, University of California, August 1, 1952. Manuscript, without revision, requested by the Editor of this Journal, February 4, 1954.

on the determination of the minimum steady magnetic field required to change the direction of magnetization of samples having polarizations that have been proved to have remained stable. The results should be valid for an upper limit if the ferromagnetic ingredient consists of a single magnetic phase with no memory and if the direction of magnetization is not controlled by grain shape anisotropy. We know that the latter is not the case in several instances from studies of directional susceptibility, but the samples used in the preliminary studies showed the behavior of a system consisting of more than one phase.

A summary plot (Fig. 1) shows all the directions of magnetization we have been able to measure in undeformed flat-lying Paleozoic sediments. These come from 14 exposures in the northeastern United States (see Fig. 2; not included in original

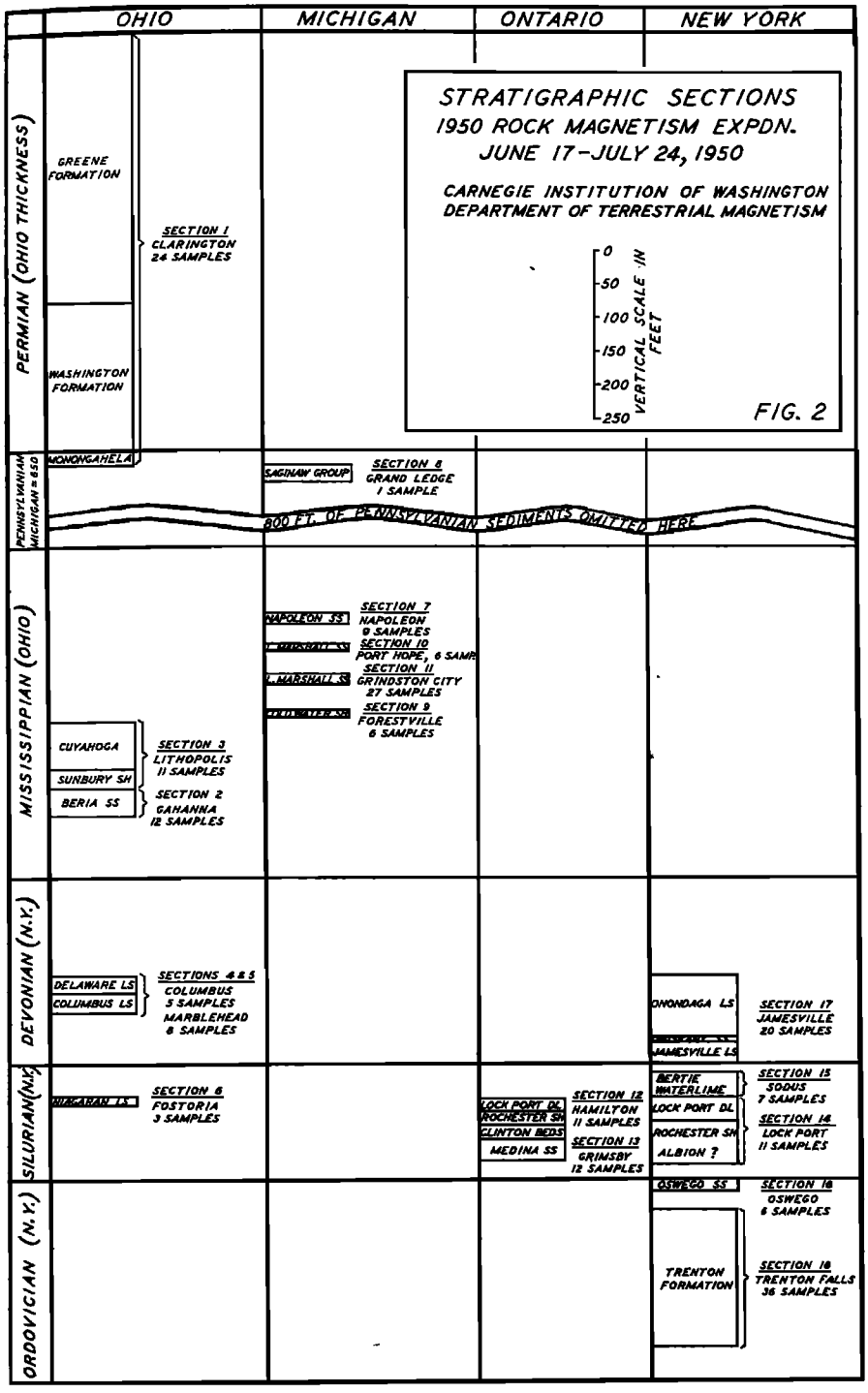


POLARIZATIONS OF FLAT-LYING PALEOZOIC SEDIMENTS

presentation). We estimate the probable experimental error of the plotted directions as not exceeding 10° in practically all cases; more than half the samples have sufficiently strong moments that the total probable error for them does not exceed 5°. In addition to these 182 measured samples, there were about 50 samples with moments too weak to measure. The exposures range in age from Ordovician through Permian, and somewhat more emphasis has been given the older sections. It is significant to note that all of these polarizations except one (open circle in Fig. 1) are north seeking downward. Likewise, it is significant that there is a preponderance of polarizations having high inclination. The distribution of polarizations by octants is: NE 23 per cent, SE 22 per cent, SW 18 per cent, and NW 37 per cent. This distribution does not change greatly when the coordinates are shifted so as to make one of the division planes pass through the present direction of the earth's field.

Faced with these observations, the question next arises whether the observed

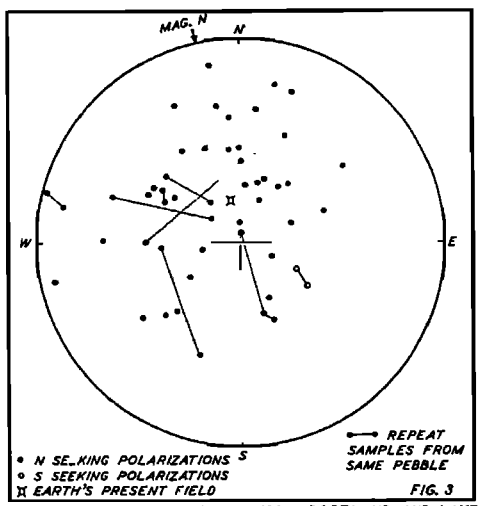
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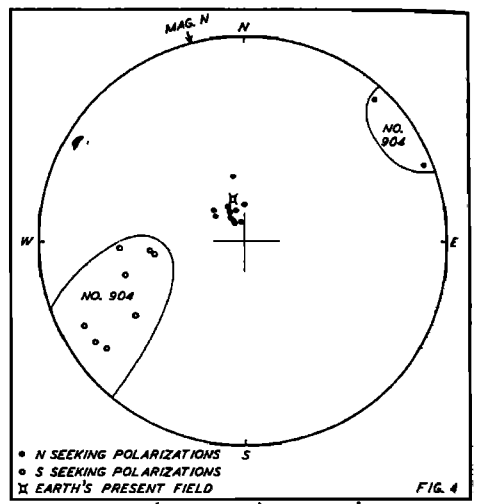
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directions of magnetization give a correct picture of the fields that prevailed at the time the samples were deposited. On two grounds, we must necessarily question each observed polarization: magnetic instability may allow the polarizations to turn gradually from some initial direction toward some averaged direction of the earth's field, and there may be mechanical deflections of the primary magnetization (distortion of the bed during drying and compaction, non-ideal settling of the particles, etc.).

In the matter of the magnetic stability of rocks, we can now make a number of statements: we can offer some excellent evidence that certain specific rocks have faithfully retained their magnetizations since at least late pre-Cambrian time; a number of Ordovician sediments from the Hudson Valley-Lake Champlain region have obviously had their primary magnetizations completely obliterated; we can show examples of late pre-Cambrian, Ordovician, Devonian, Mississippian, and Triassic rocks which clearly are neither completely stable nor unstable (example in Fig. 3). The tests of stability we have made by observing the present directions of magnetization in ancient conglomerates and distorted beds. Completely stable polarizations in conglomerates are observed today as vectors at random; partially stable polarizations are biased preferentially in some direction, and we consider that polarizations are completely unstable when, from pebble to pebble in the



UPPER TRIASSIC CONGLOMERATES, PORTLAND AND LAKE QUONNIPAUG, CONN.; SUFFERN, N.Y.



CHAZY BRECCIA (ORDOVICIAN), REYNOLD'S POINT, ISLE LA MOTTE, VERMONT

conglomerate, they are oriented in the same direction. This type of observation, of course, tells nothing directly about the cause of instability, nor about the conditions or time interval during which the polarizations went through their greatest change.

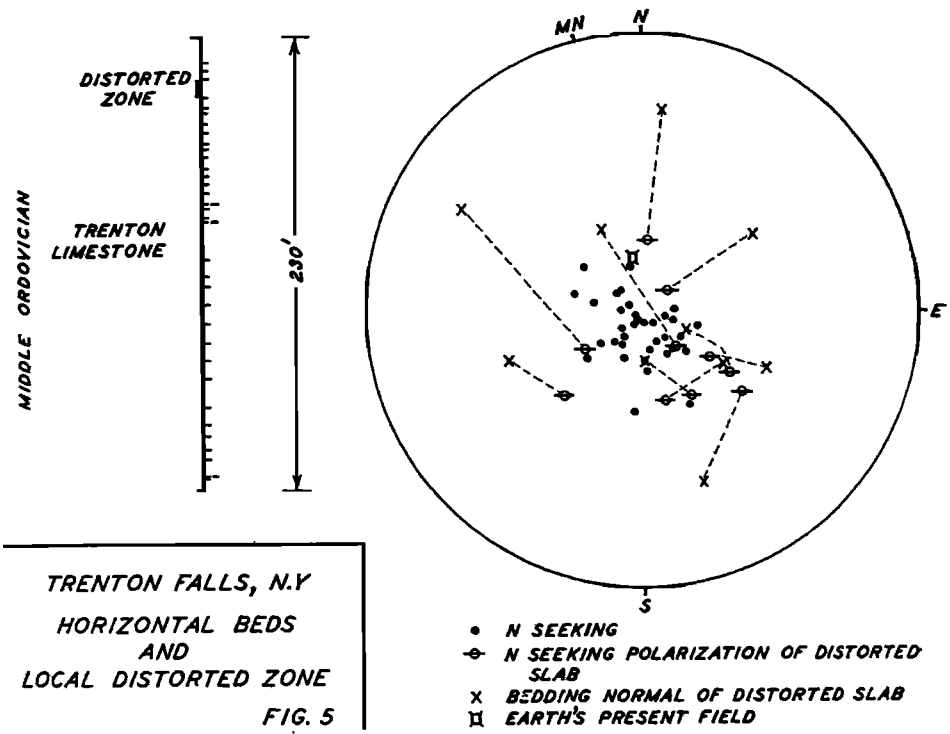
In one Ordovician conglomerate (Fig. 4), we have observed that all rocks do not have the same stability. Of 12 nearly-contiguous pebbles sampled, 11 had the same direction of magnetization within 10°. Ten cores from the twelfth pebble had magnetizations scattering ±30° from their average direction, and this average

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direction differed from that of the other "pebbles" by 130°. Obviously, the material in this particular piece had a significantly greater degree of magnetic stability.

In our early limited experience, we found by this sort of testing that remarkable stability seemed to be the rule. You may imagine our initial disappointment when we first encountered the clearly unstable polarizations and the partially stable ones. But we now feel that even from these we can learn something of value concerning the history of the earth's field. The interesting point is that these unstabilized polarizations seem to be converging, roughly speaking, towards the direction and sense of the earth's field today. The more important fact is that, for many rocks, the "time constant" of this convergence is to be measured in some cases, at least, assuming steady rates, in terms as high as hundreds of millions of years. It was obvious at the start of the study that many magnetically unstable rocks must exist in nature; the task for us was to locate and identify at least a few with polarizations of significant stability.

An illustration of this important point that we can find examples of significant stability in the flat beds is provided by Figure 5. Here the clustering of polarizations



TRENTON FALLS, N.Y
HORIZONTAL BEDS
AND
LOCAL DISTORTED ZONE
FIG. 5

of the horizontal undeformed beds is clear. Lying significantly outside this group are the polarizations observed in samples from the local distorted zone which is composed of the same material as the enclosing flat strata. We can here recognize samples with long time constants of stability, and we observe the steeply-inclined normal polarizations of the same material in the undisturbed condition.

As to the possible causes of instability, we can make several guesses. We know,

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for example, that AC fields in the laboratory cause some samples to change polarizations rapidly, and we can wonder whether the various modulations of the earth's field spread out over long times can have the same effect. An intriguing notion is that in the equatorial regions where the earth's field has its smallest values and where high-latitude auroral disturbances are lacking, the rocks may show a greater degree of magnetic stability.

From observations in relatively young Triassic conglomerates (Fig. 3), we received the hint that heating may be of considerable importance in the matter of stability. Of course, to be dogmatic about the thermal history of any geologic section is risky, but clearly these Triassic samples had been appreciably heated by nearby intrusive diabase sills. Of the 50 samples measured, 48 are north seeking downward and scattered considerably in direction. We may wonder whether the completely unstabilized polarizations we found in the Ordovician of the Hudson Valley-Lake Champlain region are an indication of a more rugged thermal history than we at first believed, rather than a valid indication of the probable stability of sediments in general. We feel that, in comparison with all of these conglomerates, the flat-lying Paleozoic rocks probably have had milder thermal treatment, and hence it is reasonable that they may have more stable polarizations.

One other line of argument that strengthens the belief that the flat-lying Paleozoic sediments probably have a significant degree of magnetic stability is as follows: we have sampled a limited number of exposures of pre-Cambrian gneisses, diabases, and ore bodies, and to summarize the observations quickly, we will say that the observed residual magnetic moments are greatly scattered in direction—whether the sample-to-sample distance be on the scale of inches, feet, or miles. Certainly, we are dealing here with ferromagnetic material having a high degree of stability. It is presumably this same sort of material which is responsible for the magnetic moments of many of the sediments after it is ground to small size and redeposited. One would predict that, if anything, the finer grain size produced by erosion would lead to greater stability in the sediments.

We believe that in yet another way we can argue in favor of a significant degree of magnetic stability in the flat-lying Paleozoic sediments. Occasionally, we have encountered individual beds where in a matter of inches the directions of magnetization change as much as 150°. We have come into the habit of regarding these "wild" polarizations as stable ones which received their dispersion by mechanical disturbance taking place at the time of deposition or possibly shortly thereafter.

This interpretation, though reasonable, may not be justified, for we have no direct observational evidence under controlled conditions of the influence of grain size and shape, compaction, turbulent deposition, dessication, etc., on polarizations. Perhaps in this respect our studies are particularly lacking, for we undoubtedly are overly naive in assuming that at the time of deposition all the sediments we sample acquired a magnetization that reflects accurately the prevailing earth's field (neglecting stability considerations). That this assumption is inaccurate seems to be illustrated by the steady decrease of inclination with increase of age in the Pleistocene varved clays. It is only reasonable to expect that the same processes affected the Paleozoic sediments, but to what extent we do not know.

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To review our stand on the flat-lying Paleozoic beds, we feel we can make a fair case for believing in a significant degree of magnetic stability among certain sediments, and yet we cast doubts on the hypothesis that at the time of deposition the sediments invariably acquire a valid primary magnetization that they retain without change in the face of the effects of compaction, distortion, diagenesis, etc. With what, then, are we left?

Taking as pessimistic a view as possible, I think the situation is no worse than the following: at the time of deposition and shortly thereafter, the sediments have polarizations that, for a guess, are rarely more than 90° from the prevailing direction of the earth's field. Some of these polarizations are stable, and many more have time constants of stability (under the conditions of thermal history prevailing in the flat beds) which our experiments show (by biased randomness in conglomerates) are to be reckoned in terms of tens and perhaps hundreds of millions of years. We observe that these polarizations today are almost exclusively north seeking downward, and that the preponderance of them is close to the direction of the earth's field today ±30°. The clustering thus represents a sort of averaging of whatever wandering the earth's field has undergone, plus additional scatter introduced by mechanical processes and varying time constants of stability. Because of the long time constant of stability for at least a few rocks which have been identified (in flat beds and conglomerates), we reach the conclusion that for the most part the earth's field has remained in about its present orientation and sense, on the average, to give the observed pattern of polarizations. The behavior of the unstable conglomerate pebbles is consistent with this conclusion. Very short term and infrequent reversals of the earth's field would not be revealed by this system.

A few words should now be added about the complex problem of appraising the significance of inverse polarizations. In the Appalachian Mountains, we have encountered a number of widely separated sections where many dozens of feet of Silurian strata with inverse magnetizations are interbedded with others of identical appearance having normal polarizations. Through these sections, there are a number of well-established faunal zones characterized by ostracods having a very restricted occurrence in the geologic time scale. Outside the deformed geosynclinal area, the flat-lying contemporaneous beds containing the same faunal assemblages have normal polarizations exclusively. Granting the stability of all these polarizations and taking them at their face value, these findings would seem to indicate that such reversals as the earth's field may have undergone in Silurian time took place over a relatively small area of the earth that was nevertheless still a large area when expressed in square miles; they took place very rapidly; they lasted for a short time; they took place without yielding transitional directions between the two senses; and, finally, they took place in a zone that later was destined to become first deeply buried by additional sedimentation and then later become folded and deformed during a mountain building orogeny. In view of the complexity of this problem, it seems well not to believe that these findings prove a reversal of the earth's field during Silurian time.

It seems to us wiser to attempt to understand these polarizations along the lines suggested by Néel's theoretical treatment, which offers various ways in which

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reversals of the sense of magnetization may take place. Recent verification of some of this theory, at least, seems to be provided by Nagata's latest observations on the magnetization behavior of a Japanese inverse extrusive rock.

We are trying to get a better insight into this problem by studies of some excellent pre-Cambrian inverse diabase dikes in Michigan.* To date, we have established that they are uniformly magnetized to within at least 1/4″ of the contact, and that cooling the samples quickly in the earth's field from above the Curie point does not yield inverse magnetizations. We have started some annealing experiments, because we have found an isolated dike only two feet wide which has uniform inverse magnetizations. We would be quite excited if we should find that the two interacting phases called for by Néel's theory require more time to exsolve than is available in the usual rapid cooling during laboratory experiments. Predicting from this theory, one would say that the phases had sufficient opportunity to exsolve during the time of cooling of the two-foot dike.

To summarize, from a consideration of all of our observations, it is reasonable to believe that through most of Paleozoic time, except for possible brief excursions, the earth's magnetic field retained essentially its present orientation and sense. Our observations do not give adequate proof that this was so, but there are now enough facts known about inverse polarizations to make it reasonable to ask for detailed support of any proposal that they demonstrate reversals of the sense of the earth's field.

*Details have been published (J. Geophys. Res., 58, 243-260, 1953).

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